

Weston Pangle

+1 (734) - 730 - 1810 | panglew.github.io | pangle.weston@gmail.com | linkedin.com/in/panglew/

EDUCATION

University Of Michigan, Ann Arbor, MI

2021 - 2025

Bachelor's of Science in Engineering, *Computer Science Engineering*

Relevant Coursework: DS&A, Computer Organization, Software Engineering, Web Systems, Operating Systems (6 credit), Computer Networks, Game Design & Dev, Intro Machine Learning, Distributed Systems, Database Management Systems, Computer Security

Languages and Technologies: Proficient - C/C++, Python, Go, Git | Intermediate - SQL, Unity, JIRA | Familiar - AWS, Docker, JavaScript, CSS, HTML, Java, MongoDB

LEADERSHIP AND EXPERIENCE

Orientation Leader – *University of Michigan*

April 2024 – Current

- Coordinated daily orientations for 1,000–1,500 attendees by organizing schedules, managing logistics, and delivering presentations, demonstrating strong organizational skills, public speaking ability, and clear communication of information
- Facilitated small group activities, breakout discussions, and Q&A sessions, collaborating with team members to ensure smooth operations and create a comfortable, welcoming environment that encouraged student participation and engagement

Technical Game Designer – *Driver's Seat Studios: University of Michigan*

October 2024 – December 2024

- Contributed to team's 1st place finish across 14 games at a competitive showcase by designing dynamic level layouts, implementing interactive gameplay systems, and ensuring smooth player progression for a polished, engaging experience
- Formulated and built unique, memorable interactive level designs and object scripts that enhanced gameplay mechanics, balanced challenge and pacing, and maximized player immersion through environmental storytelling and responsive systems
- Sourced, integrated, and customized high-quality assets, including advanced particle systems, fire and water shaders, and stylized cel shading via custom Shader Graphs, while optimizing visual fidelity and maintaining strong performance across devices

ML Engineer – *Michigan Data Science Team: University of Michigan*

September 2022 – January 2023

- Developed a Pokémon "auto-battler" using Python and supervised learning techniques, achieving 85% success in battle predictions and optimizing performance with a move recommendation algorithm based on Pokémon types, moves, and traits
- Collaborated with a team to design, train, and refine machine learning models for strategic battle simulations, significantly enhancing the AI's ability to predict and adapt to battle scenarios, leading to improved gameplay dynamics and outcomes

PERSONAL PROJECTS

Sharded Key-Value Service w/ Paxos Groups | Distributed Systems | Go

April 2025

- Engineered a distributed, fault-tolerant key/value store with Paxos-based replication and dynamic sharding across replica groups, ensuring high availability and consistency even during group joins, failures, and network partitions
- Designed and implemented a Shard Master service to handle consistent, serialized reconfiguration, enabling smooth and reliable shard reassignment during group joins and failures, and maintaining system stability under high load
- Achieved linearizable client operations with at-most-once semantics, ensuring minimal data loss and improved system performance during stateful shard migration, while effectively managing concurrent reconfigurations and reducing downtime

Reliable Transport Protocol | Computer Networks | C++

November 2024

- Developed a custom reliable transport protocol over UDP, ensuring in-order, reliable data delivery with comprehensive handling for packet loss, delay, corruption, duplication, and reordering, significantly improving network reliability
- Implemented a sliding window mechanism for wSender & wReceiver to effectively manage packet transmission and acknowledgement, incorporating retransmission timers and checksum validation to enhance protocol robustness
- Optimized data transfer efficiency by modifying wSender & wReceiver to support selective acknowledgement and retransmission, reducing unnecessary data retransmissions and significantly improving overall network throughput and performance

Network File System | Advanced Operating Systems | C++

April 2024

- Engineered a multithreaded network file system, utilizing a manager-worker threading model to optimize handling of concurrent file operations, significantly boosting system throughput and reducing latency in high-load scenarios
- Designed and implemented thread-safe file and directory operations with mutexes and upgradeable locks, ensuring strong data integrity while minimizing performance bottlenecks during simultaneous access and improving system responsiveness
- Conducted extensive testing and validation of system performance across various operational states, ensuring robust compatibility with both initialized and new file systems while maintaining stability and reliability under heavy client load

ADDITIONAL

More Projects: HTTP Proxy w/ DNS, Thread Library, Instagram Clone, Search Engine, Metroid Remake, SQL & NoSQL Database

Interests: Building Computers, Playing Video Games, Weightlifting, Basketball, Photography, Collecting Cologne